

Name: Monisha Raghini Chelle

Roll No:2520030537

Subject: OSSP

Pipes

1.

Code:

```
#include <stdio.h>

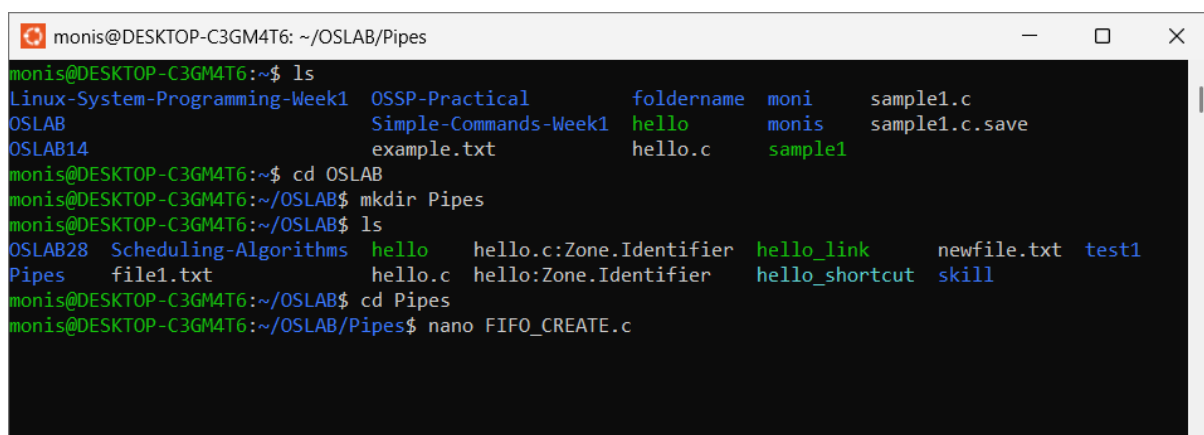
#include <sys/types.h>

#include <sys/stat.h>

int main()
{
    if (mkfifo("myfifo", 0666) == -1)
    {
        perror("mkfifo");
        return 1;
    }

    printf("FIFO created\n");

    return 0;
}
```



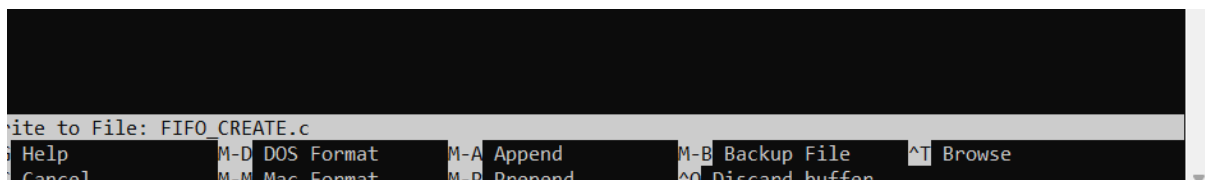
```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni    sample1.c
OSLAB                          Simple-Commands-Week1  hello      monis   sample1.c.save
OSLAB14                        example.txt          hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ mkdir Pipes
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd Pipes
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ nano FIFO_CREATE.c
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
GNU nano 8.7.1 FIFO_CREATE.c *
#include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>

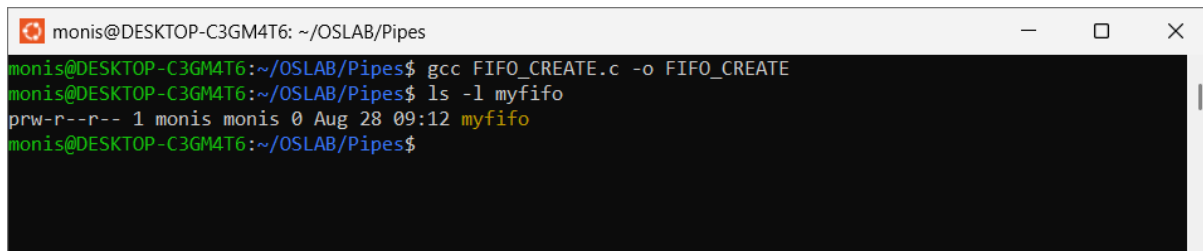
int main()
{
    if (mkfifo("myfifo", 0666) == -1)
    {
        perror("mkfifo");
        return 1;
    }

    printf("FIFO created\n");

    return 0;
}
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSLAB  OSLAB14
OSLAB28  Scheduling-Algorithms  Pipes
OSLAB  Simple-Commands-Week1  file1.txt
OSLAB14  example.txt  hello.c
OSLAB28  hello.c:Zone.Identifier  hello.c:Zone.Identifier
Pipes  file1.txt  hello.c  hello:Zone.Identifier
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ mkdir Pipes
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes  file1.txt  hello.c  hello:Zone.Identifier  hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd Pipes
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ nano FIFO_CREATE.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ gcc FIFO_CREATE.c -o FIFO_CREATE
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ ls -l myfifo
ls: cannot access 'myfifo': No such file or directory
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ ./FIFO_CREATE
FIFO created
```

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes\$'. The first command is 'gcc FIFO_CREATE.c -o FIFO_CREATE'. The second command is 'ls -l myfifo'. The output of the second command is 'prw-r--r-- 1 monis monis 0 Aug 28 09:12 myfifo'. The prompt returns to 'monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes\$'.

1. Compile the program

```
gcc FIFO_CREATE.c -o FIFO_CREATE
```

2. Check before running

```
ls -l myfifo
```

Output:

ls: cannot access 'myfifo': No such file or directory

Mention/write:

Before executing the program, myfifo does not exist.

3. Execute the program

```
./FIFO_CREATE
```

Output:

FIFO created

Mention/write:

The program creates a FIFO named myfifo.

4. Check again after execution

```
ls -l myfifo
```

Output will look similar to:

```
prw-r--r-- 1 monis monis 0 ... myfifo
```

Mention/write:

After executing the program, myfifo is created successfully. The p at the beginning indicates that it is a FIFO (named pipe).

So your complete practical sequence is:

Compile:

```
gcc FIFO_CREATE.c -o FIFO_CREATE
```

Before execution:

```
ls -l myfifo
```

→ No such file or directory

Execute:

```
./FIFO_CREATE
```

→ FIFO created

After execution:

```
ls -l myfifo
```

→ myfifo is displayed

2. FIFO_READ

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("myfifo", O_RDONLY);

    read(fd, buffer, sizeof(buffer));

    printf("Received: %s\n", buffer);

    close(fd);

    return 0;
}
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
GNU nano 8.7.1 FIFO_READ.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

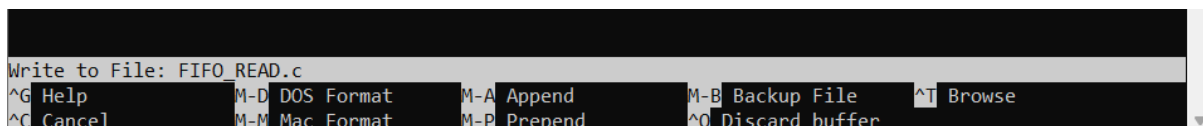
    fd = open("myfifo", O_RDONLY);

    read(fd, buffer, sizeof(buffer));

    printf("Received: %s\n", buffer);

    close(fd);

    return 0;
}
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ nano FIFO_READ.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ gcc FIFO_READ.c -o FIFO_READ
```

Don't run yet ..
Create FIFO_write.c
And Compile

3. FIFO_write.c

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd;
    char message[] = "Hello from Writer";

    fd = open("myfifo", O_WRONLY);

    write(fd, message, strlen(message) + 1);

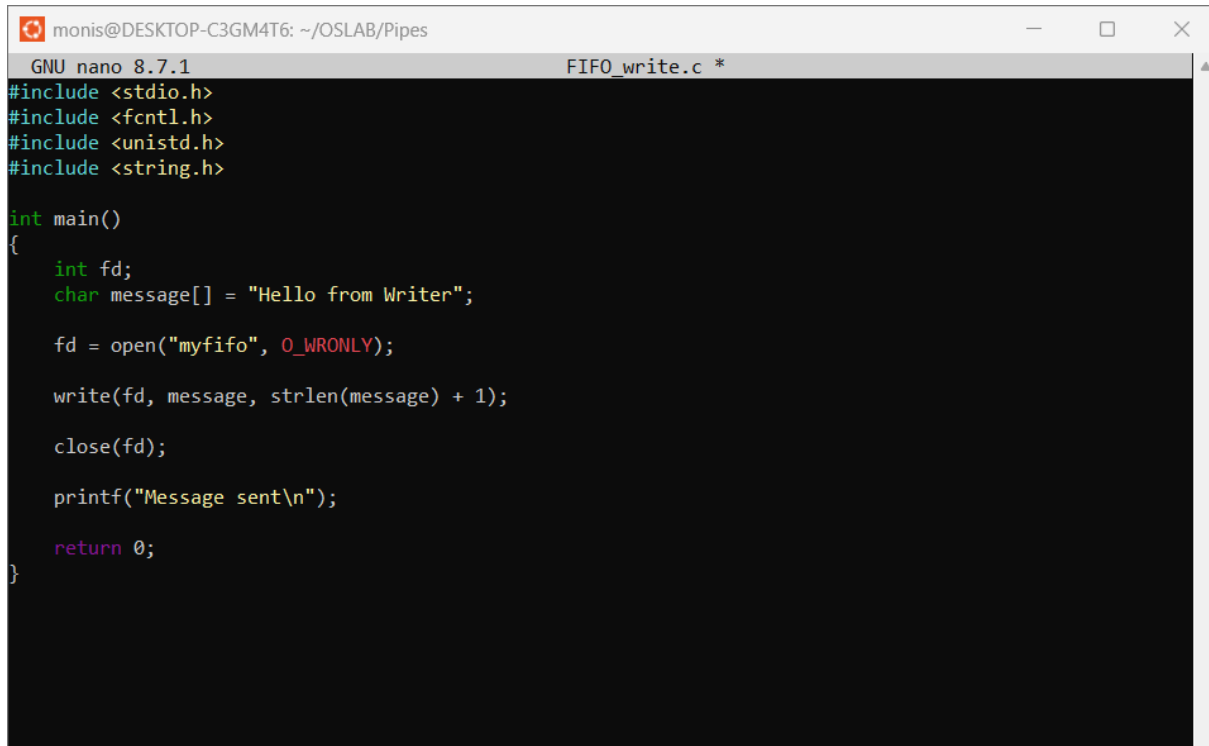
    close(fd);
}
```

```

    printf("Message sent\n");

    return 0;
}

```



```

monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
GNU nano 8.7.1 FIFO_write.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd;
    char message[] = "Hello from Writer";

    fd = open("myfifo", O_WRONLY);

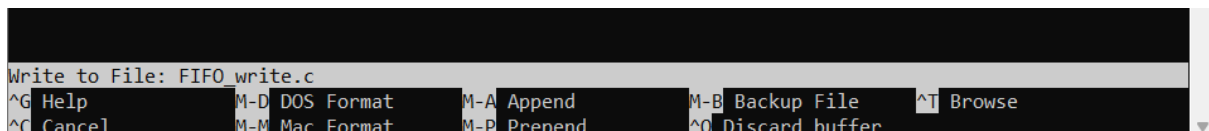
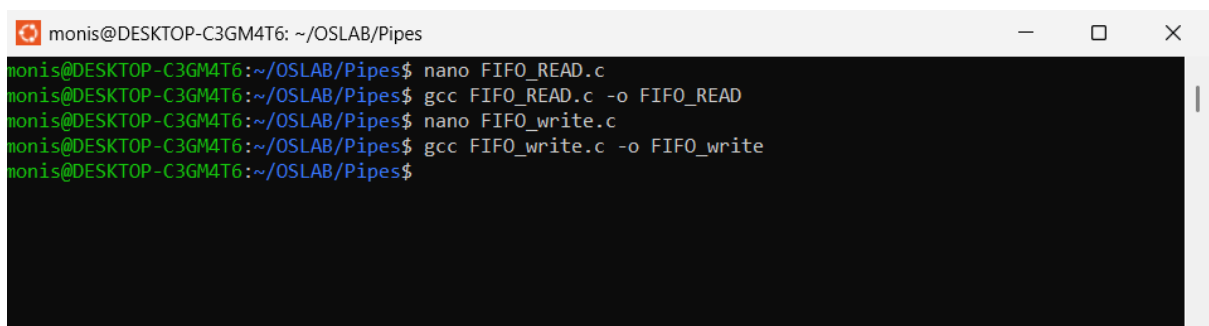
    write(fd, message, strlen(message) + 1);

    close(fd);

    printf("Message sent\n");

    return 0;
}

```

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ nano FIFO_READ.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ gcc FIFO_READ.c -o FIFO_READ
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ nano FIFO_write.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ gcc FIFO_write.c -o FIFO_write
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$

```

Open New Terminal:
 Now we have 2 Terminal
 Terminal 2

```
monis@DESKTOP-C3GM4T6: ~$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Pipes
Linux-System-Programming-Week1  OSLAB14      Simple-Commands-Week1  foldername  hello.c  monis    sample1.c
OSLAB                          OSSP-Practical  example.txt          hello      moni     sample1  sample1.c.save
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello    hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd Pipes
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ ls
FIFO_CREATE  FIFO_CREATE.c  FIFO_READ  FIFO_READ.c  FIFO_write  FIFO_write.c  myfifo
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ ls -l
total 60
-rwxr-xr-x 1 monis monis 16048 Aug 28 09:13 FIFO_CREATE
-rw-r--r-- 1 monis monis  220 Aug 28 09:07 FIFO_CREATE.c
-rwxr-xr-x 1 monis monis 16136 Aug 28 09:16 FIFO_READ
-rw-r--r-- 1 monis monis  253 Aug 28 09:16 FIFO_READ.c
-rwxr-xr-x 1 monis monis 16176 Aug 28 09:18 FIFO_write
-rw-r--r-- 1 monis monis  292 Aug 28 09:17 FIFO_write.c
prw-r--r-- 1 monis monis    0 Aug 28 09:12 myfifo
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ gcc FIFO_CREATE.c -o FIFO_CREATE
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ ls -l FIFO_CREATE
-rwxr-xr-x 1 monis monis 16048 Aug 28 09:24 FIFO_CREATE
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ ./FIFO_CREATE
mkfifo: File exists
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$ ls -l myfifo
prw-r--r-- 1 monis monis 0 Aug 28 09:12 myfifo
monis@DESKTOP-C3GM4T6:~/OSLAB/Pipes$
```

Now both terminal
1 & 2

```
monis@DESKTOP-C3QMT6: ~/OSLAB/Pipes
monis@DESKTOP-C3QMT6:~$ ls
Linux-System-Programming-Week1  OOSP-Practical  foldername  moni  sample1.c
OSLAB                           Simple-Commands-Week1  hello  monis  sample1.c.save
OSLAB14                          example.txt      hello.c  sample1
monis@DESKTOP-C3QMT6:~$ cd OSLAB
monis@DESKTOP-C3QMT6:~/OSLAB$ mkdir Pipes
monis@DESKTOP-C3QMT6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt                  hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3QMT6:~/OSLAB$ cd Pipes
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ls
FIFO_CREATE  FIFO_CREATE.c  FIFO_READ  FIFO_READ.c  FIFO_WRITE  FIFO_WRITE.c  myfifo
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ls -l
total 60
-rwxr-xr-x 1 monis monis 16048 Aug 28 09:13 FIFO_CREATE
-rw-r--r-- 1 monis monis 220 Aug 28 09:07 FIFO_CREATE.c
-rwxr-xr-x 1 monis monis 16136 Aug 28 09:16 FIFO_READ
-rw-r--r-- 1 monis monis 253 Aug 28 09:16 FIFO_READ.c
-rwxr-xr-x 1 monis monis 16176 Aug 28 09:18 FIFO_WRITE
-rw-r--r-- 1 monis monis 292 Aug 28 09:17 FIFO_WRITE.c
prw-r--r-- 1 monis monis 0 Aug 28 09:12 myfifo
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ gcc FIFO_CREATE.c -o FIFO_CREATE
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ./FIFO_CREATE
FIFO created
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ls -l myfifo
prw-r--r-- 1 monis monis 0 Aug 28 09:12 myfifo
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ gcc FIFO_READ.c -o FIFO_READ
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ gcc FIFO_WRITE.c -o FIFO_WRITE
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ gcc FIFO_WRITE.c -o FIFO_WRITE
Message sent
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$
```

```
OSLAB  OOSP-Practical  example.txt  hello  moni  sample1  sam
pl01.c.save
monis@DESKTOP-C3QMT6:~$ cd OSLAB
monis@DESKTOP-C3QMT6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt                  hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3QMT6:~/OSLAB$ cd Pipes
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ls
FIFO_CREATE  FIFO_CREATE.c  FIFO_READ  FIFO_READ.c  FIFO_WRITE  FIFO_WRITE.c  myfifo
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ls -l
total 60
-rwxr-xr-x 1 monis monis 16048 Aug 28 09:13 FIFO_CREATE
-rw-r--r-- 1 monis monis 220 Aug 28 09:07 FIFO_CREATE.c
-rwxr-xr-x 1 monis monis 16136 Aug 28 09:16 FIFO_READ
-rw-r--r-- 1 monis monis 253 Aug 28 09:16 FIFO_READ.c
-rwxr-xr-x 1 monis monis 16176 Aug 28 09:18 FIFO_WRITE
-rw-r--r-- 1 monis monis 292 Aug 28 09:17 FIFO_WRITE.c
prw-r--r-- 1 monis monis 0 Aug 28 09:12 myfifo
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ gcc FIFO_CREATE.c -o FIFO_CREATE
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ls -l FIFO_CREATE
-rwxr-xr-x 1 monis monis 16048 Aug 28 09:24 FIFO_CREATE
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ./FIFO_CREATE
mkfifo: File exists
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ls -l myfifo
prw-r--r-- 1 monis monis 0 Aug 28 09:12 myfifo
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ./FIFO_READ
Received: Hello from Writer
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$
```

```
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ./FIFO_WRITE
Message sent
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$
```

```
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$ ./FIFO_READ
Received: Hello from Writer
monis@DESKTOP-C3QMT6:~/OSLAB/Pipes$
```